

IME775 Assignment 1 (Chapters 1-2)

Foundations of machine learning and linear algebra

Student Name: _____

Due Date: _____

Instructions

- Show your work for all calculations.
 - You may use Python for arithmetic, but you must show formulas and steps.
 - Round final numeric answers to 3 decimal places unless otherwise stated.
 - Submit a single PDF or Markdown file.
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Part A: Concepts from Chapter 1 (Short Answer)

1) In 3-5 sentences, define machine learning as described in Chapter 1 and explain how it differs from traditional programming.

2) For the "cat brain" model, write the linear model equation using inputs hardness (x_0), sharpness (x_1), weights w_0 , w_1 , and bias b .

Then describe how the output is converted into the three decisions: run away, ignore, approach and purr.

3) Give one example each of:

- A classification problem
 - A regression (quantitative estimation) problem
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Part B: Vectors and Geometry (Chapter 2)

Let $a = [3, 4]$ and $b = [-2, 1]$.

- 4) Compute the L2 norm of a and b .
 - 5) Compute the dot product $a \cdot b$.
 - 6) Compute the angle between a and b (in degrees).
 - 7) Are a and b orthogonal? Explain using a numeric check.
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Part C: Cosine Similarity and Documents

Document feature vectors (machine, learning, biology):

- $d1 = [5, 2, 0]$
- $d2 = [3, 4, 0]$
- $d3 = [0, 1, 8]$

- 8) Compute cosine similarity for all pairs: $(d1, d2)$, $(d1, d3)$, $(d2, d3)$.
 - 9) Which pair is most similar? Explain in 2-3 sentences.
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Part D: Linear Combinations and Dependence

Let $v = [7, 3]$, $u1 = [1, 0]$, $u2 = [0, 1]$.

- 10) Find scalars α_1 and α_2 so that $v = \alpha_1 * u1 + \alpha_2 * u2$.

Let $w1 = [2, 4]$, $w2 = [1, 2]$, $w3 = [3, 5]$.

11) Determine whether $\{w_1, w_2, w_3\}$ is linearly dependent or independent. Provide a short justification (1-3 sentences) and show a dependence relationship if dependent.

Part E: Matrices and Transposes

Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix}$

$B = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix}$

- 12) Compute AB .
13) Is BA defined? If yes, compute BA . If no, explain why.
14) Verify the transpose property $(AB)^T = B^T A^T$ using your computed matrices.
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Part F: Linear Transformations and Orthogonality

15) Consider the rotation matrix for 45 degrees:

$$R = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

with $\theta = \pi/4$ and vector $x = [1, 0]$.

- 15a) Compute Rx .
15b) Show that R is orthogonal by checking $R^T R = I$.
15c) Show that the length of x is preserved by rotation.
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Part G: Eigenvalues and Eigenvectors

Let

$$A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$$

- 16) Compute the eigenvalues of A.
- 17) For each eigenvalue, provide one (non-zero) eigenvector.